

MIC Sound Intensity Collection

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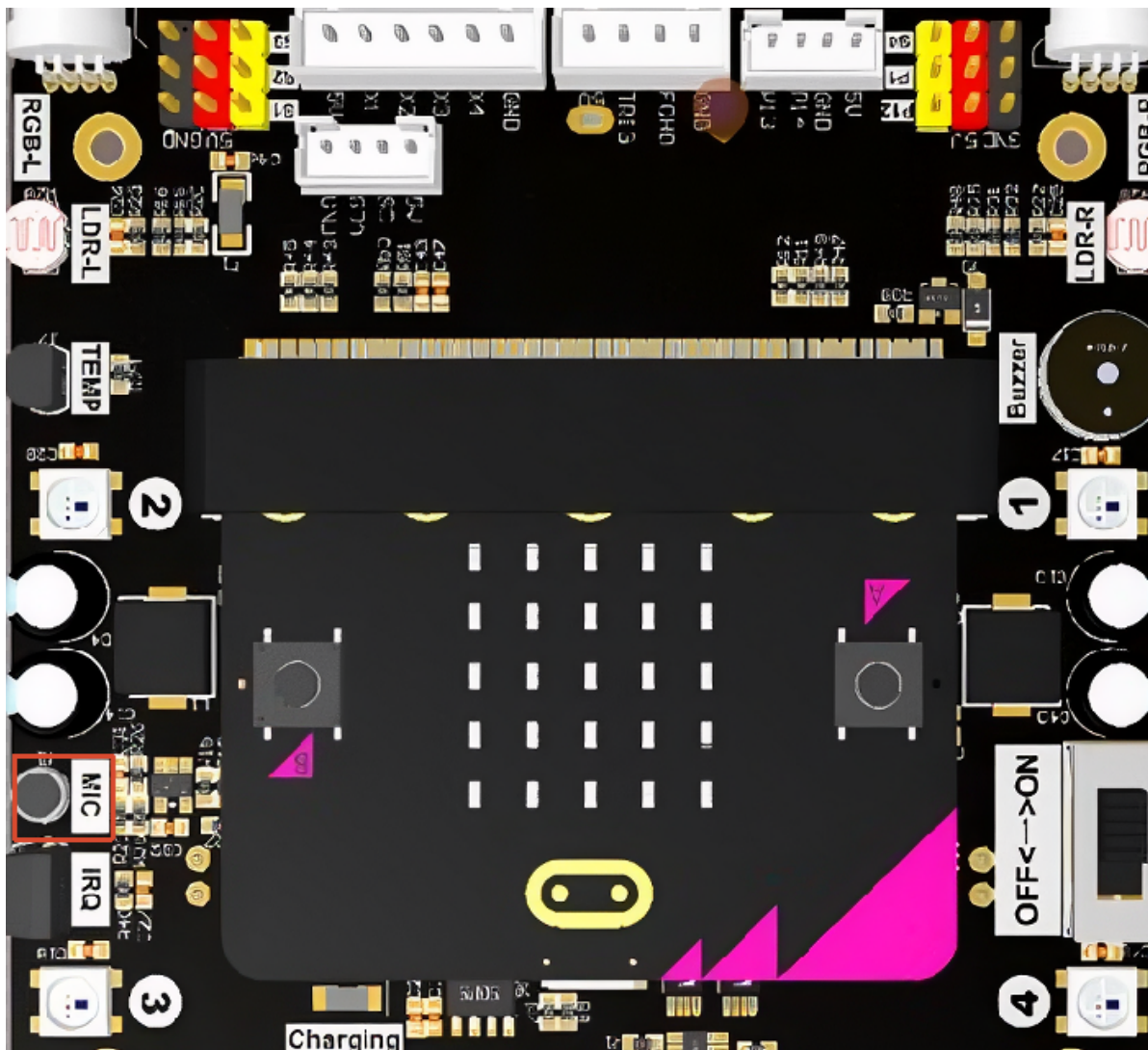
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Experimental Purpose

The Lastbit car is equipped with a sound sensor, marked with the silkscreen MIC. In this section, we will learn how to read the intensity information returned by the sound sensor. The larger the value, the noisier the environment the car is currently in.

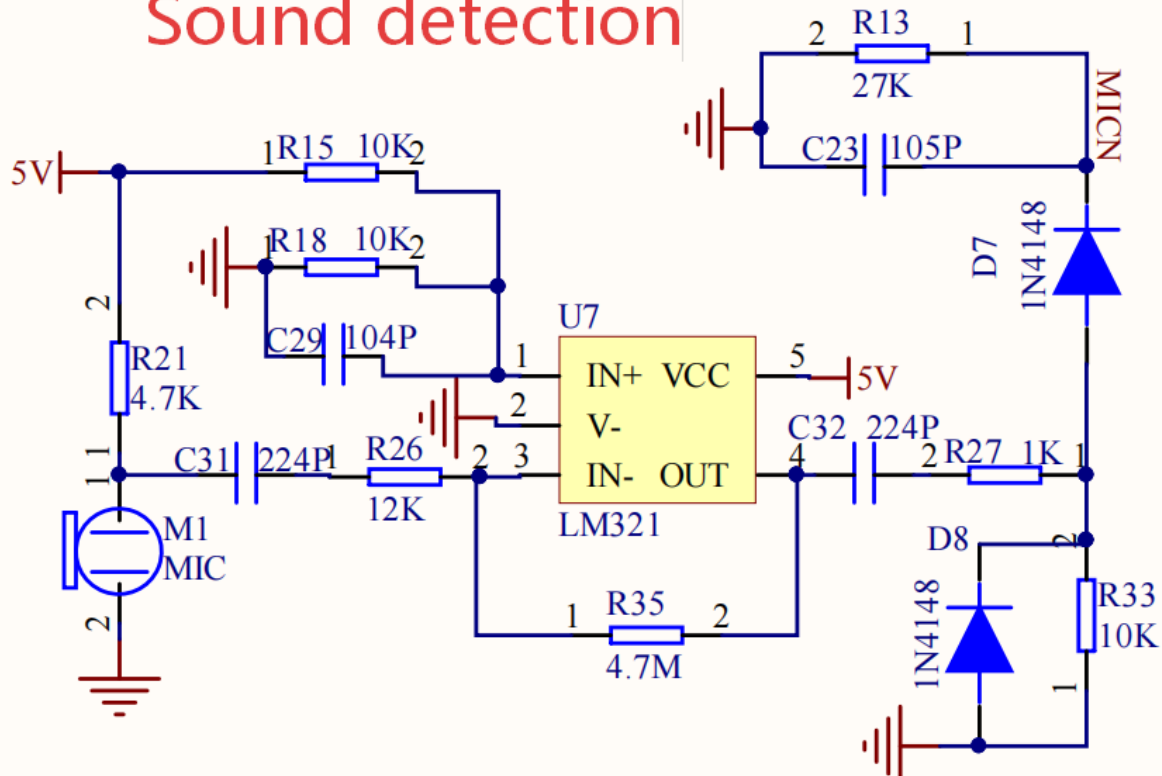
[!NOTE]

The Micro:bit V1.0 version does not have a built-in microphone, and sound detection can be implemented through the MIC on the expansion board.



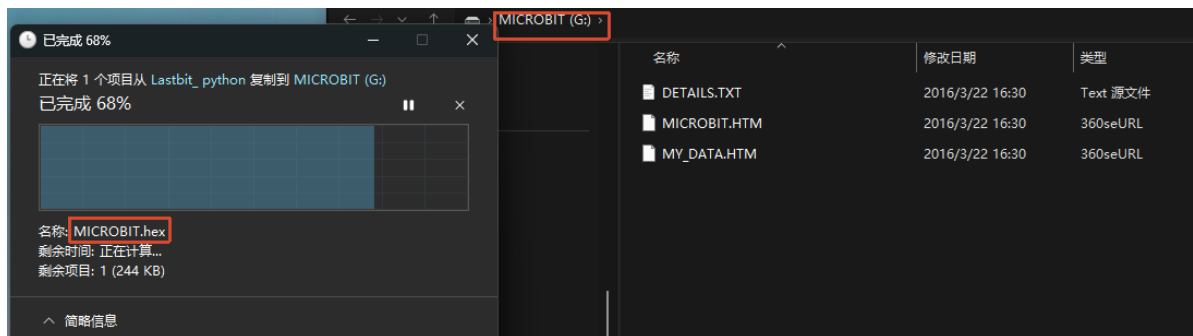
The onboard sound sensor consists of an electret microphone (commonly known as a mic head), which, together with signal amplification and clamping protection, forms a complete sound detection circuit.

Sound detection



Experimental Steps

Before flashing, make sure `LASTBIT.hex` has been flashed; otherwise, an error will occur and the module cannot be imported.



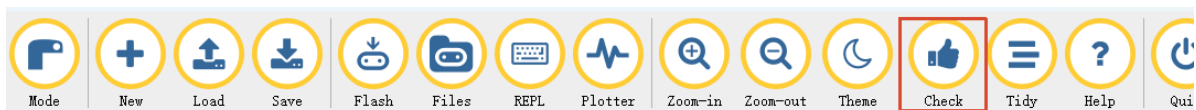
1. Before flashing, switch the switch to the OFF position and connect the computer to the microbit main board using a microUSB cable. **Note: it should be the Microbit interface, not the expansion board's Charging port.**
2. Open the Mu software. Here you can directly copy the program source code we provide; the operation steps are as follows. You can also enter the code yourself in the editor window. Note! All English characters and symbols should be entered in English input mode. Use the Tab key for indentation, and end the last line with a blank line.
Click the `New` button in the toolbar at the top left, and a file titled untitled will appear.



3. Copy the content of the `Code Implementation` section provided below into the current file. You will notice a small red dot after the title bar, indicating that the code content has been modified but is not yet saved.

Whether or not you save does not affect the code check and flashing.

4. At this point, you can click `Check` to check whether the code format has any problems.



5. If you have connected the MICROBIT to the PC in advance following the steps above and have flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently, and the editor will show `Copied code onto micro:bit.` at the bottom, indicating that the flashing is complete. After flashing, the indicator light will no longer blink.



6. Unplug the Microusb data cable and switch the switch to the ON position.

Code Implementation

```
# 6.1 MIC_Sound_Intensity_Collection(Expansion board MIC)

from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

while True:

    # Read sound data
    sound_data = Lastbit.sound_detect()
```

```
if sound_data is not None:

    # Read sound intensity
    volume = sound_data[2]
    display.scroll(str(volume))

    # Print sound intensity
    # print(str(volume) + '\n')

sleep(100)
```

`sound_detect(min, max, avg)`: reads the sound intensity detected by the MIC. It returns a tuple of three values (minimum, maximum, average), among which the average is the most suitable for judgment.

Experimental Phenomenon

The Microbit LED matrix will display the sound intensity data obtained by the sound sensor.